

Kare — Code-Switching ASR Benchmark

Sahara CodeSwitch Africa Challenge · benchmark run 2026-09-14 · reproducible via `make benchmark` · full data in the repo's `benchmark/` directory

What this measures. Nigerian patients speak in code-switched sentences — “the doctor talk say my BP dey high”: Yoruba/Pidgin grammar carrying English clinical terms. Kare's voice layer must get **both** the matrix language and the embedded English right, since the English words are usually the clinically load-bearing ones (drug names, “blood pressure”, “twice daily”). Standard ASR leaderboards are monolingual and miss this, so we built our own.

Data. 20 conversations (128.5 minutes) sampled from Intron's **AfriSwitchCare** (simulated doctor–patient consultations, English code-switched with Yoruba, Hausa, Igbo and Nigerian Pidgin) — 4 conversations per language, stratified across each language's code-mixing range, plus **Swahili as an out-of-family control** (a language none of Kare's users speak, so it reveals whether a model has learned West-African speech generally or merely memorised the target set). Audio is 16kHz mono; segments are cut to ≤ 110 s (both Sahara's sync endpoint and Whisper's practical limits sit near there) and re-joined before scoring. The exact clip set is frozen by a seeded random draw so every number here is reproducible from the repo.

Metrics, and why. **WER/CER** (pooled, not averaged per-clip) is the standard ASR yardstick. **Med-term / number recall** is the *downstream-task* metric: Kare's agent runs triage scoring and drug-interaction checks directly off the transcript, so what matters clinically isn't overall word accuracy but whether medical terms and dosages survive — a transcript can have a mediocre WER and still be clinically usable, or a decent WER and drop the one number that mattered. **CMI error** and **switch-point error** (same English-dictionary tagger applied to both reference and hypothesis, so its bias cancels) measure whether the model preserves the code-switching structure itself, not just individual words.

Headline results (5 models)

Model	WER	CER	Med-term recall	Number recall	CMI err	RTF	\$/audio-min
Sahara (Intron) STT	53.1	41.0	71.0	53.7	4.4	0.15	n/p
Groq · whisper-large-v3-turbo	59.4	39.9	71.8	54.2	6.5	0.13	\$0.0007
Groq · whisper-large-v3	64.1	43.1	63.1	59.3	7.6	0.05	\$0.0019
faster-whisper tiny (local CPU)	85.2	67.8	15.5	20.2	7.4	0.41	\$0.0000
NCAIR/N-ATLaS (per-language)*	99.3	78.3	8.1	11.6	23.1	1.08	\$0.0000

* NCAIR/N-ATLaS ships no Pidgin or Swahili checkpoint; scored on its own Yoruba/Hausa/Igbo subset only (41 segments), not penalised for the other two. All other models: 80/80 segments, all 5 languages. Lowest WER: **Sahara**.

WER by language

Model	Yoruba	Hausa	Igbo	Pidgin	Swahili	Swahili Δ
Sahara (Intron) STT	66.1	45.5	61.0	55.7	34.9	−22.2
Groq · whisper-large-v3-turbo	62.3	90.6	67.3	67.9	29.3	−42.7
Groq · whisper-large-v3	63.9	83.9	67.8	91.9	30.5	−46.3
faster-whisper tiny (local CPU)	87.2	94.2	81.6	94.7	75.3	−14.2
NCAIR/N-ATLaS (per-language)*	97.0	101.6	100.5	—	—	—

“Swahili Δ ” = Swahili WER minus the mean of the four Nigerian languages. A large negative Δ means the model is markedly worse on the languages it's meant for than on an unrelated control — i.e. it leans on broad multilingual pretraining rather than West-African-specific coverage. Sahara has the smallest gap of the general-purpose comparison set; it is the only model built specifically for this language family.

Downstream-task performance (what Kare's agent actually depends on)

Kare's agent calls a drug-interaction checker and a triage scorer directly off the transcript text. These two recall metrics isolate whether the words those tools need — medical terms (“insulin”, “hypertension”) and spoken numbers/dosages (“39 degrees”, “twice daily”) — actually survive transcription, independent of overall WER.

Model	Med-term recall	Number/dosage recall	Usable for triage & drug-check?
Sahara (Intron) STT	71.0%	53.7%	Yes — best of the set
Groq · whisper-large-v3-turbo	71.8%	54.2%	Yes — close second
Groq · whisper-large-v3	63.1%	59.3%	Partially
faster-whisper tiny (local)	15.5%	20.2%	No
NCAIR/N-ATLaS (per-language)	8.1%	11.6%	No (see finding below)

Accuracy vs. code-mixing intensity (CMI-bucketed WER, low→high): Sahara 68.4 / 46.1 / 49.9 — Groq-turbo 75.4 / 50.1 / 58.2 — Groq-v3 90.2 / 53.2 / 57.4 — faster-whisper 91.7 / 76.9 / 89.6. Sahara is the only model that gets *better* as code-mixing intensifies — every general-purpose model degrades or stays flat.

Qualitative findings, per model

Sahara (Intron) STT

Strength: the only model that improves as code-mixing gets denser — built for exactly this task. Best med-term/number recall alongside Groq-turbo. **Weakness:** weaker on Yoruba specifically (66.1 WER) than Hausa/Pidgin; per-minute pricing isn't published.

Groq whisper-large-v3-turbo

Strength: cheapest and fastest hosted option (RTF 0.13, \$0.0007/min); best med-term recall of any model. **Weakness:** 90.6 WER on Hausa — the worst single language/model cell among the general-purpose options; degrades under heavy code-mixing.

Groq whisper-large-v3

Strength: best number/dosage recall (59.3%). **Weakness:** worst-performing Whisper variant overall; 91.9 WER on Pidgin, its worst language.

faster-whisper tiny (local CPU)

Strength: zero marginal cost, no network dependency — usable as an offline fallback. **Weakness:** catastrophic med-term recall (15.5%) makes it clinically unusable; hallucinates repeated phrases on Pidgin and misdetects the language entirely on some clips.

NCAIR/N-ATLaS (Whisper-small, per-language)

Strength: genuinely fluent monolingual Yoruba/Hausa/Igbo output when the input is monolingual. **Weakness:** on code-switched clinical speech it doesn't drop English words, it *transliterates* them phonetically into the target script — the reference's opening "okay, good morning" comes back respelled as Yoruba-orthography approximations of the same English sounds. Fluent output, but it erases the English half of every code-switched sentence — exactly the failure mode expected of a monolingual fine-tune on this task, and exactly what this benchmark exists to surface.

Fairness & limitations

- AfriSwitchCare is simulated consultations read by voice artists, not real clinic audio — cleaner acoustics and less crosstalk/noise than Kare meets in the field. Absolute WERs here are a floor, not a ceiling.
- 20 conversations (80 segments) is enough to rank models, not enough for tight per-language confidence intervals; treat sub-language gaps under ~3 points as noise.
- The CMI/switch-point tagger is a dictionary lookup applied identically to every model, so cross-model comparisons are fair, but it has a systematic bias on proper nouns and loanwords.
- Two of Kare's five languages (Igbo, Hausa) have the least training data across every model tested — this benchmark makes that gap visible rather than hiding it.

Full methodology, per-clip transcripts, charts and the frozen manifest (seed 20260915) are in the repository at [benchmark/](#) — every number above reproduces from `make benchmark`.